

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and delivered **geotechnical analysis** for foundation systems and earthworks on major infrastructure projects, contributing to a **15% reduction** in preliminary design iteration cycles.
- Prepared comprehensive **design reports**, technical specifications, and site walkovers for industrial and commercial developments, ensuring adherence to project scope and client requirements.
- Managed and interpreted **geotechnical ground investigations**, providing critical technical support during the construction phase through periodic site visits and data analysis.
- Integrated **digital tools and automation** into project workflows, enhancing data processing efficiency for slope stability analyses by approximately **20%**.
- Developed project management skills by coordinating with cross-functional teams and supporting senior engineers in client requirement gathering for new commissions.
- Contributed to the implementation of **sustainability principles** within engineering solutions, focusing on material optimization and environmental impact assessments for earthworks.

SKILLS

Geotechnical Engineering – Geotechnical Design, Foundation Design, Retaining Structures, Slope Stability, Earthworks, Soil Mechanics, Ground Investigations

Software & Tools – AutoCAD, GIS, Numerical Analysis Software, Python, R

Reporting & Documentation – Design Reports, Technical Specifications, Desk Studies, Site Walkovers

Project Management – Planning, Supervision, Coordination, Client Liaison, Stakeholder Management

Domain Knowledge – Infrastructure Development, Civil Engineering, Structural Engineering, Sustainability in Engineering

PROJECTS

Foundation Design for a European Data Centre

- Performed detailed **geotechnical design and analysis** for deep and shallow foundations supporting a large-scale hyperscale data centre in Europe.
- Interpreted **ground investigation data**, including CPT and borehole logs, to inform design parameters and mitigate risks associated with varied subsurface conditions.
- Generated comprehensive **design reports and drawings** that were integrated into the overall construction plan, supporting an estimated **10% improvement** in construction sequencing efficiency.

Slope Stability Analysis for Irish Infrastructure

- Conducted **slope stability analyses** using numerical modelling software for a significant Irish infrastructure project, assessing potential failure modes.
- Developed automated scripts in **Python** to process raw survey data, reducing the time spent on data preparation for analysis by roughly **30%**.
- Collaborated with senior engineers to refine model inputs based on site-specific conditions and provided technical input for risk mitigation strategies.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)